

Design and Implementation of a Grid Connected Single Phase Inverter for Photovoltaic System

Md. Jahangir Hossain, Md. Raqibull Hasan, Monowar Hossain and Md. Rafiqul Islam
Department of Electrical and Electronic Engineering; Khulna University of Engineering & Technology;
Khulna, Bangladesh
E-mail: jhossain2k2@gmail.com, rafiq043@yahoo.com

Abstract—This paper reports the design procedure and performance evaluation of an improved quality microcontroller based sine wave inverter for grid connected photovoltaic (PV) system. The power interfacing element between the PV energy and electrical grid is the inverter. The electrical energy injected into the grid depends on the amount of power extracted from the PV system and the efficient processing of this power by the inverter. The grid and PV energy synchronization is the challenge of designing the grid connected inverter. The above threats are eliminated by designing microcontroller based control circuits and considering feedback from electrical grid. The implemented inverter demonstrates that it is capable for auto synchronization and satisfactory performance for grid connected PV system.

Keywords—Photovoltaic; microcontroller; grid connected inverter; modified sine wave; synchronization and control.

I. INTRODUCTION

It is the burning question to meet the consumers' growing energy demand without creating any harmful impact on the environment. The environment pollution is occurring due to continuously burning of fossil fuels. Moreover, the fossil fuels with limited resources are running out and the security of electricity supplies is under threat. Sustainable energy sources are required to meet the electricity demand to overcome the threat of energy security. Renewable energy comes forward to solve the above problem. Among all renewable energy sources solar energy is the largest and available all over the world. PV is the direct process of converting solar energy into electricity [1-2]. It is the most potential way, due to the availability, simplicity, lower maintenance and reliable operation [3]. The main applications of PV systems are in either off grid or grid connected configurations. Off grid PV generation systems are attractive as indispensable electricity source for remote areas [4]. However, challenge to utilize solar energy from off grid PV system is comparatively high capital cost, lower efficiency, larger capacity of storage battery and more operation and maintenance cost [5-6] than grid connected configuration.

Many research works are carried out currently focusing on optimization of PV systems [4, 7-9]. The grid connected system may be the best way to optimize PV energy. The PV panel generates direct current whereas electrical grids require alternating. Thus, to utilize the PV energy as grid connected configuration, DC/DC converter; DC/AC converter and their controllers are importantly necessary. The dynamics of the PV

system is completely different from that of the conventional generator, though both of them have almost identical P-Q characteristics [10]. For these reasons special types of inverter need to utilize solar energy in grid connected system. There are several topologies to design grid connected inverter such as pulse width modulation, multilevel, modified technique etc. Among them modified sine wave (MSW) inverter topology brings some advantages such as it is more economical, easy to implement and do not interference with nearby audio signal or communication networks due to low frequency switching.

The grid connected systems are installed in those areas where a robust grid is present, and able to accept energy feeding from the PV system [11]. Therefore, system synchronization is badly needed to interface PV system with electrical grid. Several research works have been done to implement grid connected inverter [12-14]. Most of them are simulation level, some are implemented but they are complex and costly [15-16], some of them make interference with nearby audio signals. To solve the above problems one of the microcontroller (Atmel89C51) has programmed to generate MOSFET operating gate signals and the other microcontroller programmed to match with the voltage level, wave shape and controlled the different operational characteristics of the inverter. The frequency of microcontroller generated signal is four times grater than grid frequency. The inverter will generate three levels voltage wave shape. In this research work design, development and performance analysis of grid connected inverter for PV systems are clearly described. The simulation and experimental results prove that the performance of the proposed inverter is better, can reduce the cost and give a reliable support to the grid.

II. DESIGN OF PROPOSED INVERTER

The configuration of the grid connected PV system illustrates in Fig.1, which consists of a dc-bus filter, DC-DC converter, a half-bridge DC-AC converter, an output filter and system controller. The half-bridge inverter contains of two active switches and two dc voltage that can process real power bidirectionally. In addition, the inverter performs current harmonics elimination to improve power quality. The dc bus filter suppresses dc-link voltage fluctuations and filters out ac components on the dc side for maximum power point tracking, while the output filter serves as an interface between the inverter and the utility.

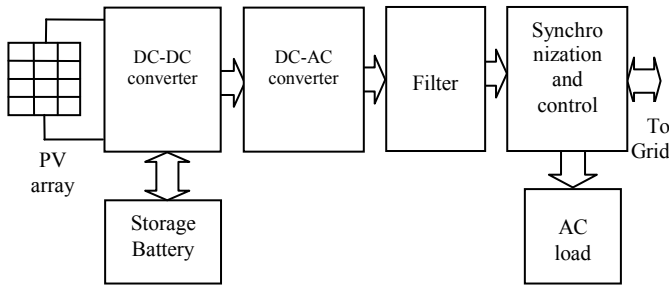


Fig.1 Block diagram of grid connected PV system

A. Methodology

The microcontroller programmed in such a way that it can generate suitable gate pulse that can operate the active switches. The source terminals of two MOSFET are connected to the low voltage terminal of center tap transformer as shown in Fig.2, center of that transformer are connected to the battery positive terminal, gates and drain terminals are connected to the microcontroller and battery negative terminal respectively. According to the microcontroller gate signal MOSFET turns on and off. The operating sequences of MOSFET are shown in Table 1. The microcontroller generates gate pulse adding some operating delay to match the grid frequency. The frequency of the designed inverter can be calculated by the following equation [17].

$$f = \frac{1}{T}, f = \frac{1}{20 \text{ ms}} = \frac{1}{0.02 \text{ s}} = 50 \text{ Hz}$$

The operating delay time can be calculated by following ways: Let microcontroller delay = x, from Fig.6 one frequency = 4x, from 50Hz frequency system one cycle = 0.02 second = 4x, one second require = 4x / 0.02 = 200x, microcontroller delay = microcontroller clock frequency / system frequency; 200x = 12000000/50, x=1200.

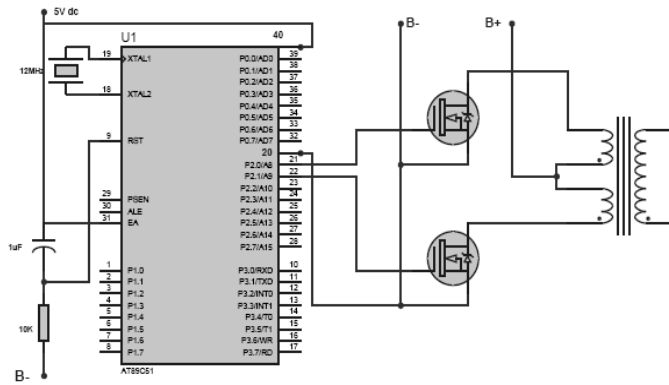


Fig.2 Signal and switching circuit for the inverter

Table.1 MOSFET Switching Sequences

Step	MOSFET 1	MOSFET 2
1	ON	OFF
2	OFF	OFF
3	OFF	ON
4	OFF	OFF

B. Hardware Design

The circuit diagram of the proposed grid connected single phase sine wave inverter using modified technology is shown in Fig. 4. It is composed of a half bridge dc-ac converter. One microcontroller (Atmel89C51) used for MOSFET operating signal for inverter and another used for control and synchronization of the inverter. The switching MOSFET (STP55NF06L) used for its excellent performance. The combination of MOSFET and microcontroller signal generates three levels a modified sine wave. Transformer and filter circuit make it pure sine wave shape. The comparator IC (LM324) compares different voltages and send signal to the microcontroller to operate switching relays via transistor (BC547) so that it can operate properly. Different voltage sources are designed to need for utilization and control scheme for both photovoltaic system and electrical grid.

C. Software Development

The flowchart of the microcontroller programming is shown in fig.3. One of the microcontroller sense grid frequency then it calculates operating delay and inverter frequency and compares it with grid frequency. The microcontroller generates gate pulse according to working sequences of MOSFET. The other microcontroller compares phase voltage, wave shape and controls the switching signals.

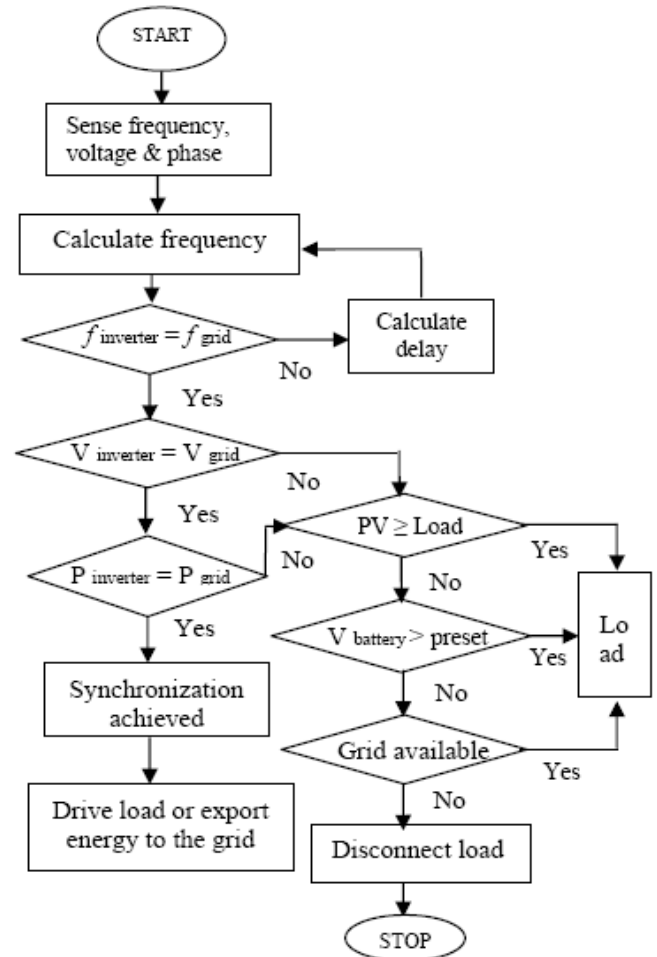


Fig.3 Flowchart of the grid connected PV inverter

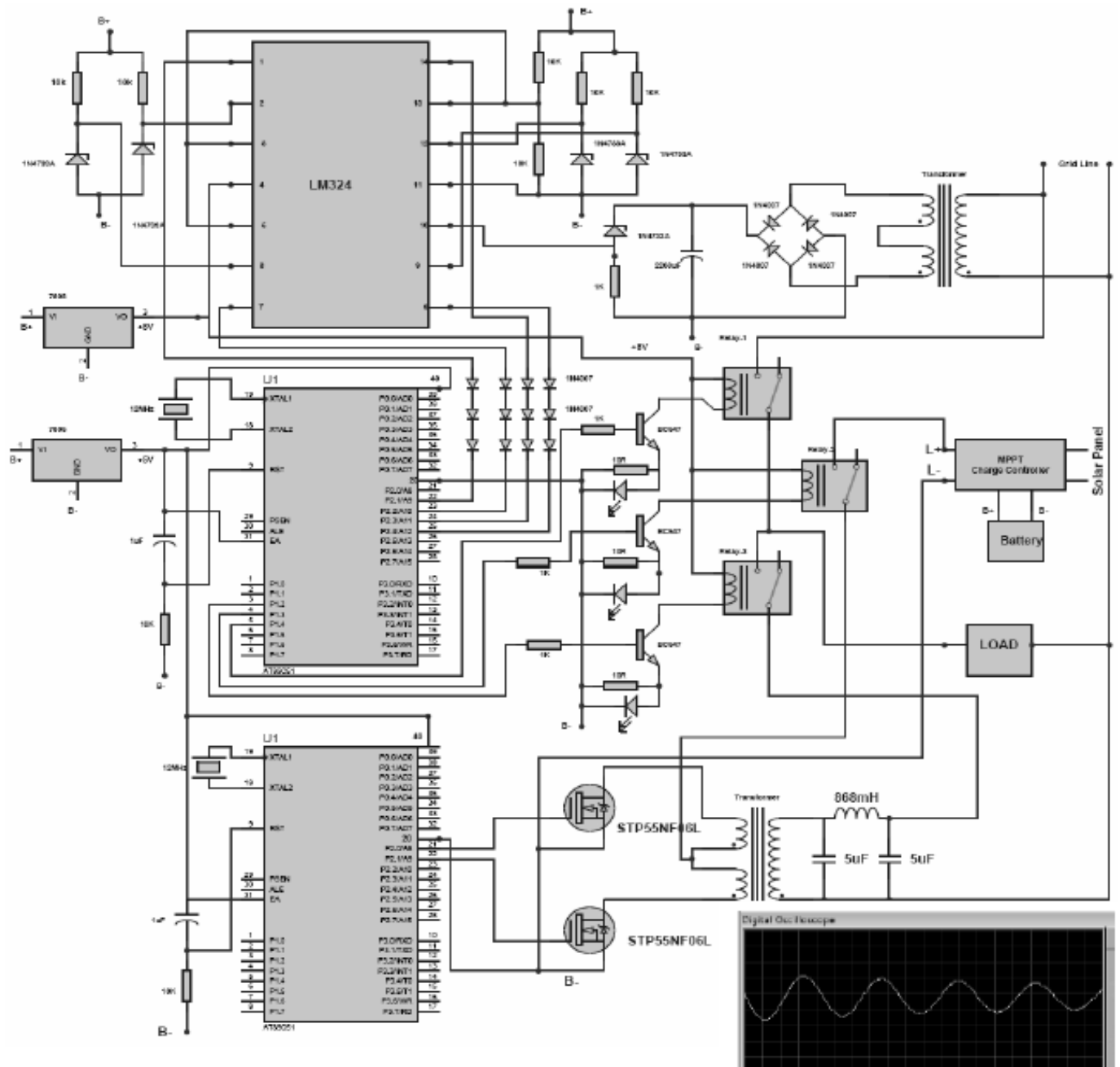


Fig.4 Proposed circuit diagram of grid connected PV inverter

D. Control Circuit and Synchronization

The control circuit continuously senses the inverter output voltage, grid voltage; PV panel (battery) terminal voltage and compares with four reference voltages by comparator. The output signals of comparator are processed by microcontroller that was initially programmed. If the battery and PV panel voltage will be in the range of 14.4V to 10.8V then only relay 2 will operate and inverter can convert dc power to ac beyond this range relay disconnect the PV panel from inverter due to over voltage and under voltage respectively. As soon as reaching the array reconnect voltage level the system

automatically connects the PV panel to the system. The control circuit checks the voltage level and phase. If the conditions of synchronization satisfy relay3 and relay1 connect PV system to the grid simultaneously. If the synchronism does not achieved feedback system change the parameters and try to synchronization. If it does not achieved and PV or backup battery is capable to drive the load then relay1 disconnect the grid from load. If PV system is not capable to drive the load then load will automatically transfer to the grid by relay 1. In this condition PV system charges the storage battery.

E. Filer Design

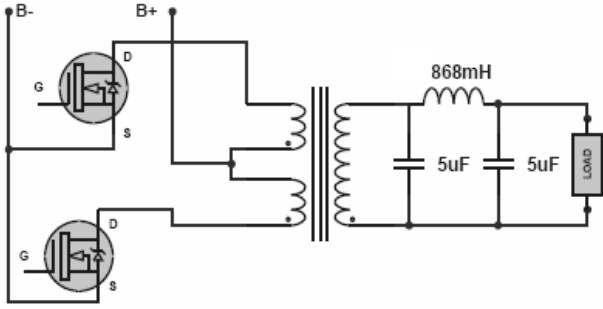


Fig.5 Filter circuit of the inverter

The filter used over here is an L-C passive π filter, consisting of an iron core inductor and non-polar capacitor. The capacitance of the capacitor is $10\mu\text{F}$ and the inductor inductance 868mH approximately.

The cutoff frequency of the combination can be determined from the equation,

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

$$f_c = \frac{1}{2\pi\sqrt{(0.868 \times 10 \times 10^{-6})}} = 54.05 \text{ Hz}$$

III. RESULT AND DISCUSSION

In this paper input output signal and set points are tested in laboratory with electrical grid, PV system and lead acid battery to protect the inverter, load and the systems. The MOSFET operating gate pulse automatically generated by microcontroller has shown in fig. 6 and 7. The switching sequences MOSFET generate modified sine wave and it feeds to the low voltage terminal of the center tap transformer (Fig.2) and it will generate almost sine wave with slide distortion has shown in Fig. 8. The distortion occurs during chopping of DC Voltage and high frequency switching of MOSFET. After filtering pure sine wave shape appears that has shown in Fig.9 and 10. Table 2 and Fig. 11 show the efficiency of the inverter at different load condition. It shows that near rated capacity efficiency is height due to less conversion losses.

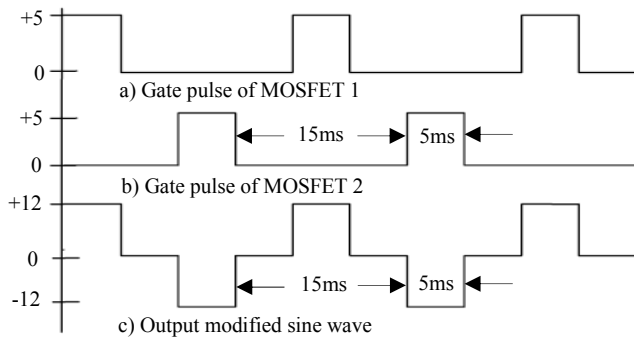


Fig.6 MOSFET gate pulse and output sequences

The proposed inverter is more economical than any other inverter topologies. It has high efficiency, low standby losses, high surge capacity and low harmonic distortion. Another advantage of this inverter is that it can generate almost perfect current or voltage waveform with low switching frequency with amplitude modulated instead of pulse width. That means the pulsating torque generated by harmonics and power losses into the machine due to this can be eliminated.

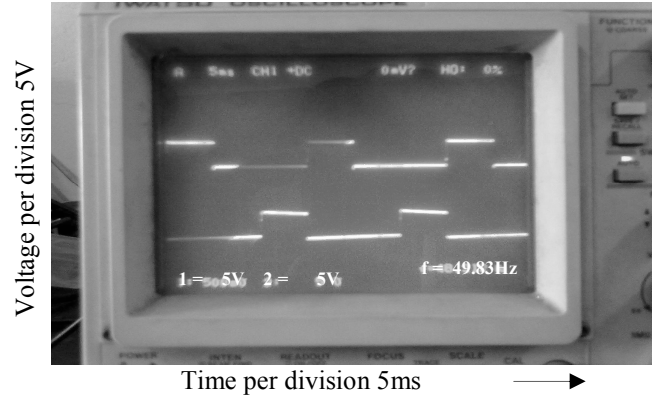


Fig.7 The MOSFET gate operating signal

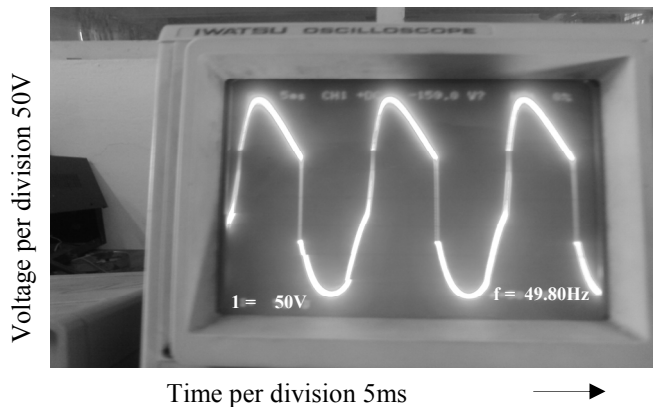


Fig.8 Output wave shape before filtering

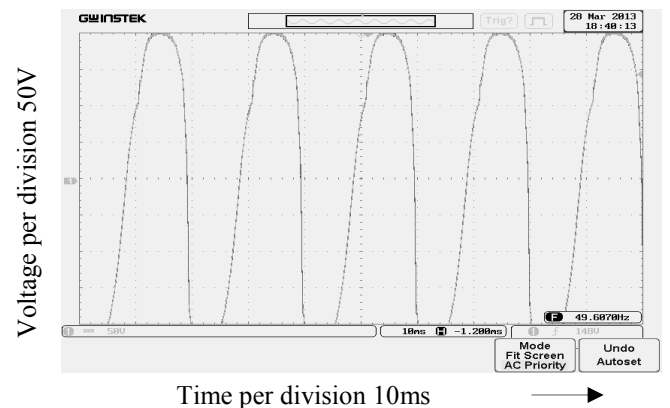


Fig.9 Output wave shape in digital oscilloscope

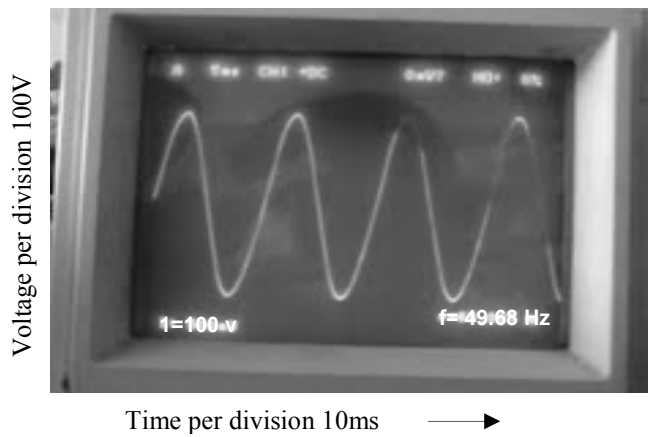


Fig.10 The output final wave shape (vol./div 100v, T/div 10ms)

Table 2: Input output data sheet

Sl. No	Input (DC)			Output (AC)			efficiency %
	V _{in} (V)	I _{in} (A)	P _{in} (watt)	V _{out} (V)	I _{out} (A)	P _{out} (watt)	
1	12	7.9	94.8	226	0.31	70.1	73.90
2	12	12.4	148.8	224	0.52	116.5	78.28
3	12	15.2	182.4	223	0.66	147.2	80.69
4	12	18.6	223.2	221	0.85	187.8	84.16
5	12	23.7	284.4	220	1.14	250.8	88.19
6	12	26.8	321.6	220	1.33	292.6	90.98
7	12	29.6	355.2	219	1.50	328.5	92.48
8	12	31.4	376.8	219	1.62	354.8	93.57
9	12	32.9	394.8	218	1.70	370.6	93.87

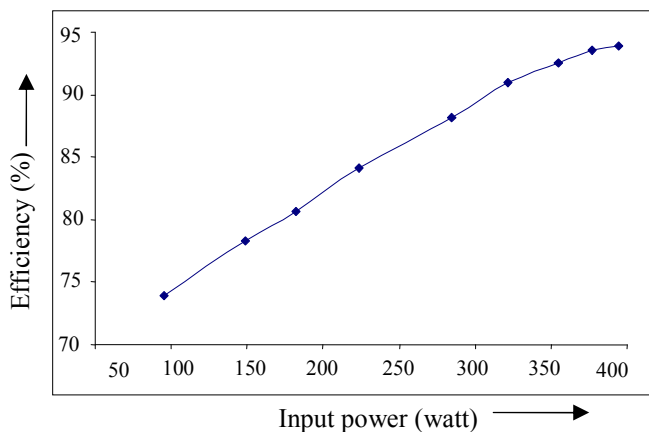


Fig.11 Efficiency-input power curve

IV. COST ANALYSIS

The inverter design and implementation cost is basically equipment and assembled cost. It can be analyzed by according to the market price of Bangladesh. The proposed 500W (93.87%) inverter equipment and assembled cost require \$43 USD. On the other hand, the same rating of

inverter (GAIA-500G, 92%) price is \$107 USD which is lowest cost in the market [18].

A. Inverter Specification

Input dc voltage	: 10.8-14.5Vdc
Input maximum dc current	: 45A _{dc}
Output ac voltage	: 200-230Vac
Output frequency	: 49-51Hz
Output maximum power	: 500W
Switching frequency	: 200 Hz

V. CONCLUSION

The proposed single phase grid connected inverter is capable of dealing power with minimum number of active switches that simplifies the system configuration and its cost is lower significantly. It does not interference with nearby audio signal or communication networks due to low frequency switching. The synchronization, switching and controlling of this inverter enhance by using microcontroller based programming that improve its quality. The inverter is able to show better performance (efficiency 93.87%). The above achievement indicates that the potentiality of proposed inverter can provide efficient and stable solution for grid connected PV system.

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